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Bereken de volgende limiet met de regel van de l'Hôpital $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan \frac{x}{2} - 1}{\cos x}$

$$\begin{aligned} & \frac{1}{\cos^2 \frac{x}{2}} \cdot \frac{1}{2} \\ \stackrel{H}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\cos^2 \frac{x}{2}} \cdot \frac{1}{2}}{-\sin x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{-\sin x \cdot \cos^2 \frac{x}{2} \cdot 2} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{-\sin \frac{\pi}{2} \cdot \cos^2 \frac{\pi}{4} \cdot 2} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} -\frac{1}{\left(\frac{\sqrt{2}}{2}\right)^2 \cdot 2} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} -\frac{1}{\frac{2}{4} \cdot 2} \\ &= -1 \end{aligned}$$

References